Remainder theorem

1 For each of the following divisors, state the value of x that needs to be substituted into $P(x)$ to find the remainder:

a $x+3$

b $8-x$

c $6-x$

d $x+10$

e $-7+6x$

f $3x-1$

g $2x+1$

h $5+4x$

2 For each of the following, find the remainder when $P(x)$ is divided by $A(x)$:

a $P(x)=-2x^4+7x^3-3x^2-6x-5,\ A(x)=x+2$

b $P(x)=3x^4-3x^3-5x^2+4x-5,\ A(x)=x+5$

c $P(x)=3x^4+5x^3-2x^2+6x+7,\ A(x)=3x+1$

d $P(x)=2x^4-3x^3+6x^2-10,\ A(x)=x-1$

e $P(x)=x^3-2x^2+9x-1,\ A(x)=x-4$

f $P(x)=4x^5-6x^3-7x^2+9x,\ A(x)=2x-5$

g $P(x)=6x^4-x^3+9x^2+10x-8,\ A(x)=4x+2$

h $P(x)=-4x^4+6x^3+4x^2-7x+7,\ A(x)=3x-1$

3 Find the value of k for each of the following:

a The remainder when $3x^3-2x^2-4x+k$ is divided by $x-3$ is 47.

b The remainder when $3x^3+4x^2+kx-3$ is divided by $x+1$ is -5.

c The remainder when $3x^3+4x^2+4x+k$ is divided by $x-2$ is 52.

d The remainder when $4x^3-2x^2+kx-1$ is divided by $x-2$ is 15.

e The remainder when $3x^3+2x^2+x+k$ is divided by $x+1$ is -3.

f The remainder when $2x^3+2x^2+kx-2$ is divided by $x+3$ is -80.

g The remainder when $2x^3-2x^2-4x+k$ is divided by $x+3$ is -57.

h The remainder when $4x^3+3x^2+kx-2$ is divided by $x+1$ is -147.

4 The polynomials $P(x)=x^3+4x^2-5x+n$ and $Q(x)=x^3+2x+17$ leave the same remainder when divided by $x+1$.
Solve for the value of n.

5 Consider the polynomial $x^{99}+1$.

a Find the remainder when the polynomial is divided by $x+1$.

Factor theorem

6 Suppose we are given a polynomial of degree n. What is the name of the process where we find two polynomials of degree less than n that multiply together to give the original?

7 What operation is the opposite of factoring?

8 When can we say that a polynomial is factored completely?

9 Define the factor theorem.

10 Show that $x+2$ is a factor of $P(x)=x^4+3x^3-7x^2-27x-18$.

11 Consider $P(x)=2x^3+8x^2+x+4$.

a Show that $x+4$ is a factor of the polynomial $P(x)$.

b Now, find the other quadratic factor.

12 Consider $P(x)=10-17x+8x^2-x^3$.

a Show that $5-x$ is a factor of the polynomial $P(x)$.

b Now, find the other quadratic factor.

13 The polynomials $4x^2-7x-15$ and $5x^2+13x+k$ have a common factor of $x+p$, where p is an integer.

a Solve for the value of p.

b Solve for the value of k.

14 Consider the following polynomials:

- $P(x)=x^4-4x^3-20x+3$
- $Q(x)=x^3-4x^2-20x+3$
- $R(x)=x^4-4x^2-20x+3$
- $S(x)=x^4-4x^3-20x^2+3$

Determine for which polynomial $(x+3)$ is a factor. Explain your answer.

15 Rewrite the following polynomials as produ[redacted] linear factors:

a $x^3 + 2x^2 - 13x + 10$

b $x^3 - 6x^2 - x + 30$

c $x^3 - 9x^2 + 23x - 15$

d $x^3 + 8x^2 + 17x + 10$

e $2x^3 - 11x^2 + 19x - 10$

f $5x^3 - 41x^2 + 83x - 15$

g $2x^3 + x^2 - 7x - 6$

h $4x^3 + x^2 - 13x - 10$